



AEO “Nazarbayev Intellectual Schools”

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CENTRE
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CHEMISTRY

Grade 12

Paper 1

May 2025

1 hour

Candidates answer on the Question Paper.

Additional Materials: Calculator
 Ruler
 Data Booklet

12CHEM/01

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

Do not use staples, paper clips, glue, or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Calculators may be used.

The total number of marks for this paper is 40.

Answer all questions in English.

For Examiner's Use	
Section A	
Section B	
Total	

This document consists of **15** printed and **1** blank page.

Section A

For
Examiner's
Use

For each question, there are four possible answers **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct by putting a tick (✓) in the correct box.

1 Which statement about mass spectrometry is correct?

- A** Mass spectrometry separates compounds.
- B** Mass spectrometry is used to determine the relative molecular mass of a compound.
- C** A mass spectrum shows the mass number plotted against abundance.
- D** Particles must be negatively charged to reproduce a mass spectrum.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

2 Which particle has the same electronic configuration as S^{2-} ?

- A** Sc^{3+}
- B** Br^{-}
- C** Se^{2-}
- D** P^{3+}

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

3 Which statement about the solubility of sulfates and hydroxides of Group 2 elements is correct?

- A** All sulfates are soluble in water, hydroxides have very low solubilities.
- B** Most sulfates are soluble in water, but barium sulfate is insoluble.
- C** Solubility of hydroxides of Group 2 elements decreases down the group.
- D** Solubility of sulfates of Group 2 elements increases down the group.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

4 Which statement(s) about the Period 3 elements is(are) correct?

- 1 The electrical conductivity increases across the metals in Period 3 from sodium to aluminium.
- 2 Going across the period, from Na^+ to Si^{4+} , the ions get smaller.
- 3 The melting point of sodium is higher than that of magnesium.

- A 1 only
- B 2 only
- C 1 and 2
- D 2 and 3

A ☐ B ☐ C ☐ D ☐

[1]

5 Ethanoic acid and ethanol are mixed together and allowed to reach equilibrium at 298 K.

The equilibrium mixture contains 1.00 mol dm^{-3} ethanoic acid and 1.75 mol dm^{-3} ethanol.

The K_c for the esterification is 4.00 at 298 K.

What is the concentration of ethyl ethanoate?

- A 2.65 mol/dm^3
- B 7.00 mol/dm^3
- C 2.00 mol/dm^3
- D 1.75 mol/dm^3

A ☐ B ☐ C ☐ D ☐

[1]

6 Which equation describes the first electron affinity?

- A $\text{Mg(g)} \rightarrow \text{Mg}^+(\text{g}) + \text{e}^-$
- B $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$
- C $\text{O(g)} + \text{e}^- \rightarrow \text{O}^-(\text{g})$
- D $\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}^-$

A ☐ B ☐ C ☐ D ☐

[1]

7 What is the shape and approximate value of bond angles in NCl_3 ?

- A Trigonal planar and 120°
- B Trigonal pyramidal and 107°
- C T-shaped and 88°
- D See-saw and 87°

A ☐ B ☐ C ☐ D ☐

[1]

8 Sulfuric acid is manufactured by the Contact process.

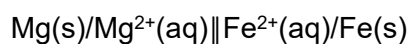
Which statement about the Contact process is correct?

- A Stage one involves oxidation of solid sulfur to sulfur(IV) oxide.
- B Stage two involves the reaction of sulfur(IV) oxide with water.
- C Stage two involves the oxidation of sulfur(VI) oxide.
- D Vanadium(V) oxide moves the equilibrium to the right.

A ☐ B ☐ C ☐ D ☐

[1]

9 Chemical energy can be converted to electrical energy in the voltaic cell below.



What is the correct equation for the spontaneous reaction?

- A $\text{Mg(s)} \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{e}^-$
- B $\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe(s)}$
- C $\text{Mg(s)} + \text{Fe}^{2+}(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{Fe(s)}$
- D $\text{Mg}^{2+}(\text{aq}) + \text{Fe(s)} \rightarrow \text{Mg(s)} + \text{Fe}^{2+}(\text{aq})$

A ☐ B ☐ C ☐ D ☐

[1]

- 10 During an exothermic reaction heat is released.



brown colourless

Which statement is correct?

- A** Increasing the temperature will cause the mixture to become darker.
B A change in temperature will not change the value of K_c .
C As the temperature increases the value of K_c will increase.
D The concentration of reactants increases as the temperature decreases.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 11 A sample of 0.80 g of magnesium reacts with excess hydrochloric acid.
 What is the volume of hydrogen, $\text{H}_2(\text{g})$, evolved measured at 25°C and $1.00 \times 10^5 \text{ Pa}$?

- A** $6.83 \times 10^{-2} \text{ dm}^3$
B $6.83 \times 10^{-5} \text{ dm}^3$
C $8.15 \times 10^{-4} \text{ dm}^3$
D $8.15 \times 10^{-1} \text{ dm}^3$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 12 Five successive ionisation energies (IE) of element **X** are shown.

First IE / kJ mol^{-1}	Second IE / kJ mol^{-1}	Third IE / kJ mol^{-1}	Fourth IE / kJ mol^{-1}	Fifth IE / kJ mol^{-1}
577	1820	2740	11600	14830

Which group of the Periodic Table contains element **X**?

- A** 1
B 2
C 3
D 4

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 13 When hot concentrated sodium hydroxide solution reacts with chlorine gas a disproportionation reaction takes place.

What are the chlorine-containing products of this reaction?

- A NaClO and NaCl
B NaClO₂ and NaCl
C NaClO₃ and NaCl
D NaClO₄ and NaCl

A ☐ B ☐ C ☐ D ☐

[1]

- 14 An element in Period 3 forms an oxide with oxidation state +4. This oxide reacts with water to produce a solution that reacts with limestone to form carbon dioxide.

Which row in the table is correct?

	Chemical behaviour of oxide	Structure of oxide
A	Amphoteric	Giant ionic
B	Acidic	Simple covalent
C	Basic	Giant covalent
D	Amphoteric	Simple covalent

A ☐ B ☐ C ☐ D ☐

[1]

- 15 Which statement describes the effect of increasing the temperature on the Maxwell-Boltzmann distribution for one mole of gaseous particles?

- A The total area under the curve changes.
B More particles have less kinetic energy.
C Activation energy of reaction remains the same.
D The number of particles with the mean kinetic energy increases.

A ☐ B ☐ C ☐ D ☐

[1]

- 16 Halides react differently with concentrated sulfuric acid at room temperature. Which statement is correct?

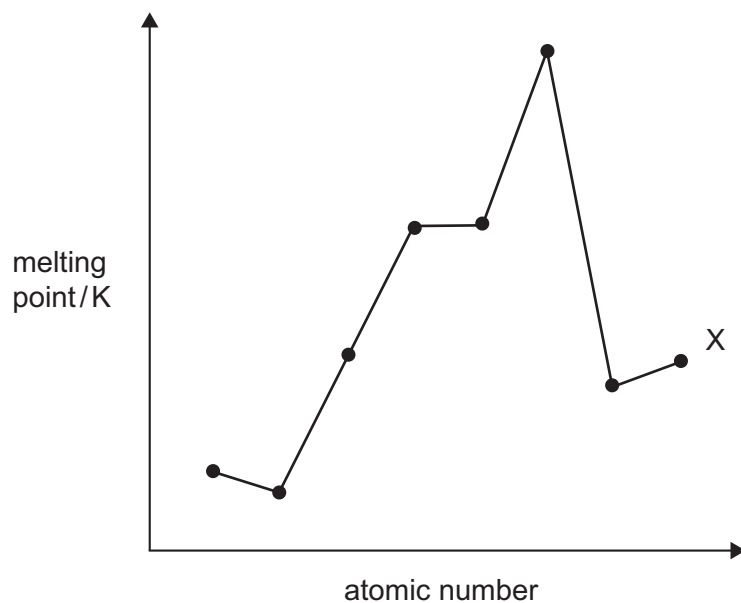
- A There is no colour change seen with sodium bromide.
B The fluoride ions reduce the sulfuric acid to sulfur dioxide.
C Reducing ability of the halide ions decreases as you go down the Group.
D No change in oxidation states with sodium chloride.

A ☐ B ☐ C ☐ D ☐

[1]

17 The graph illustrates the melting points of eight elements with consecutive atomic numbers.

What is the possible identity of the element **X**?



- A** Aluminium
- B** Carbon
- C** Phosphorus
- D** Sulfur

A ☐ **B** ☐ **C** ☐ **D** ☐

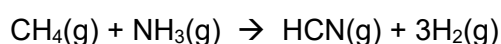
[1]

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Use

18 The table shows some enthalpy changes of reaction.

Reaction	Enthalpy change of reaction, ΔH_r / kJ mol^{-1}
$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$	-91.8
$\text{C}(\text{s}) + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g})$	-74.9
$\text{H}_2(\text{g}) + 2\text{C}(\text{s}) + \text{N}_2(\text{g}) \rightarrow 2\text{HCN}(\text{g})$	+270.3

What is the enthalpy of reaction for the reaction shown?



- A** +103.6 kJ/mol
B +255.9 kJ/mol
C +373.9 kJ/mol
D +437.1 kJ/mol

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

19 Element **X** has the largest atomic radius in its period. The oxide of **X** reacts with water. What is the acid-base behaviour of the solution produced?

- A** Strong acid
B Weak acid
C Weak base
D Strong base

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

20 Which is the correct order from highest to lowest boiling points?

- A** $\text{HCl} > \text{HBr} > \text{HI} > \text{HF}$
B $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
C $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
D $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

21 Which gas from an internal combustion engine contribute to the decrease of oxygen in human blood?

- A CO
- B N₂
- C NO
- D C_xH_y

A ☐ B ☐ C ☐ D ☐

[1]

22 How many isomers has the product of dehydrating of butan-2-ol.

- A 1
- B 2
- C 3
- D 4

A ☐ B ☐ C ☐ D ☐

[1]

23 Which statement in **NOT** correct about the mechanism of the nitration of benzene, C₆H₆.

- A The electrophile NO₂⁺ attacks positively charged carbon atom in benzene.
- B C-H bonds break heterolytically to restore delocalised ring.
- C A pair of electrons from the benzene ring is donated to the nitrogen atom in NO₂⁺.
- D Positively charged species is formed in the intermediate step.

A ☐ B ☐ C ☐ D ☐

[1]

24 Identify which compound is reduced to primary alcohol in the presence of NaBH₄.

- A CH₃COOCH₃
- B CH₃CH₂COCH₃
- C CH₃CH₂CH₂CHO
- D CH₃CH₂CH(OH)CH₃

A ☐ B ☐ C ☐ D ☐

[1]

25 A student conducts two experiments with propan-1-ol.

In the first experiment, propan-1-ol is treated with excess of acidified potassium dichromate(VI).

In the second experiment, the mixture of propan-1-ol and propanoic acid is heated in the presence of catalyst.

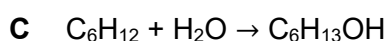
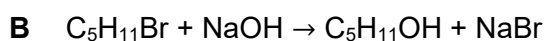
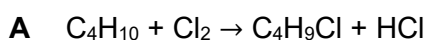
Which row in the table shows the products of these experiments?

	Experiment No. 1	Experiment No. 2
A	$\text{CH}_3\text{CH}_2\text{CHO}$	$\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3$
B	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3$
C	$\text{CH}_3\text{CH}_2\text{COOH}$	$\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3$
D	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	$\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

A ☐ B ☐ C ☐ D ☐

[1]

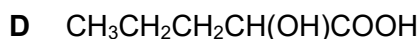
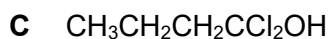
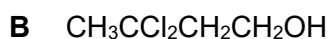
26 Which reaction has a nucleophilic addition mechanism?



A ☐ B ☐ C ☐ D ☐

[1]

27 Which compound has a chiral centre?



A ☐ B ☐ C ☐ D ☐

[1]

28 Which row describes the type of reaction and the product when 1-bromopropane reacts with dilute aqueous sodium hydroxide?

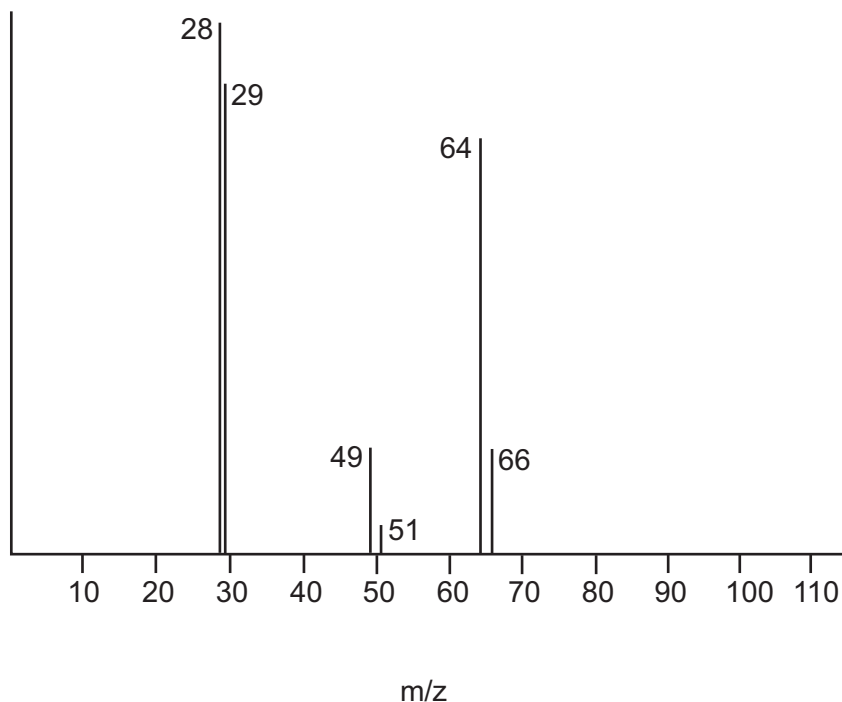
	Type of reaction	Name of product
A	Elimination	Propan-1-ol
B	Elimination	Propene
C	Substitution	Propene
D	Substitution	Propan-1-ol

A ☐ B ☐ C ☐ D ☐

[1]

29 The figure shows mass spectrum of chloroethane.

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Which peaks are responsible for $[\text{CH}_3\text{CH}_2^{35}\text{Cl}]^+$ and $[\text{CH}_3\text{CH}_2^{37}\text{Cl}]^+$ and the ratio between the peak heights?

- A 28 and 29, ratio 1:1
- B 49 and 51, ratio 3:1
- C 64 and 66, ratio 3:1
- D 49 and 64, ratio 3:1

A ☐ B ☐ C ☐ D ☐

[1]

30 Which statement describes the multiplicity and the integration value of each signal in the proton NMR spectrum of $\text{CH}_3\text{CH}_2\text{COOC}(\text{CH}_3)_3$?



- A 1 triplet, 1 quartet, 1 singlet. Ratio 3:2:9
- B 1 triplet, 1 multiplet, 1 quartet. Ratio 3:2:9
- C 1 triplet, 1 quartet, 1 singlet. Ratio 1:1:3
- D 1 triplet, 1 multiplet, 1 quartet. Ratio 1:1:3

A ☐ B ☐ C ☐ D ☐

[1]

Section B

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Use

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.
Use of the Data Booklet may be appropriate for some questions.

31 Which reagent(s) is (are) used to distinguish aldehydes from ketones.

- 1** 2,4-DNPH reagent
- 2** acidified $\text{K}_2\text{Cr}_2\text{O}_7$
- 3** Fehling's reagent

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

32 Which mixture of solutions works as a buffer solution?

- 1** $\text{CH}_3\text{COONa}(\text{aq})$ and $\text{CH}_3\text{COOH}(\text{aq})$
- 2** Excess of $\text{CH}_3\text{COOH}(\text{aq})$ and $\text{NaOH}(\text{aq})$
- 3** $\text{NH}_4\text{Cl}(\text{aq})$ and $\text{NH}_3(\text{aq})$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

33 Which molecule(s) is(are) polar?

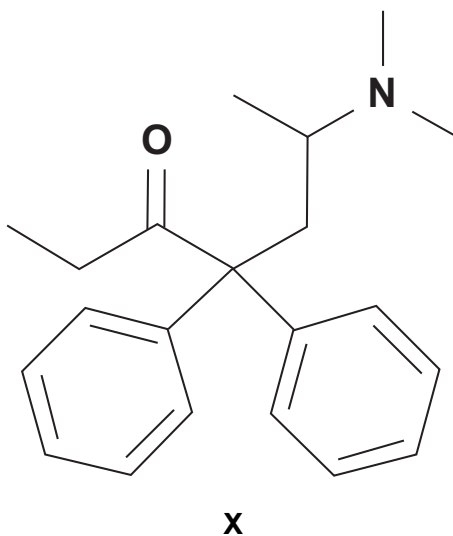
- 1** SO_2
- 2** CO_2
- 3** BF_3

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

34 What type(s) of functional groups is (are) present in compound **X**?

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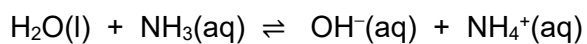


- 1 Carbonyl
- 2 Benzene
- 3 Amide

A ☐ B ☐ C ☐ D ☐

[1]

35 Water reacts reversibly with ammonia as shown in the equation.



Which species is/are acting as Bronsted-Lowry base(s)?

- 1 $\text{H}_2\text{O}(\text{l})$
- 2 $\text{NH}_3(\text{aq})$
- 3 $\text{OH}^-(\text{aq})$

A ☐ B ☐ C ☐ D ☐

[1]

36 Which solution(s) has/have a hydroxide ion concentration of $0.200 \text{ mol dm}^{-3}$?

- 1 pH=13.30, KOH
- 2 pH=13.30, $\text{Ba}(\text{OH})_2$
- 3 pH=12.70, NaOH

A ☐ B ☐ C ☐ D ☐

[1]

37 Which statement(s) is/are correct?

- 1 The higher the charge of the metal in the complex ion, the more acidic the solution.
- 2 Complex ions with 3+ metal ions release carbon dioxide from carbonates.
- 3 Hydroxonium ion is produced in reaction between metal aqua ion and water.

A ☐ B ☐ C ☐ D ☐

[1]

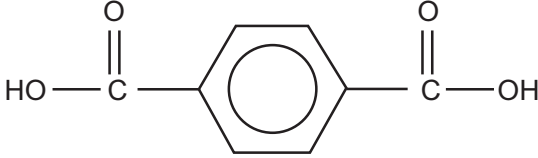
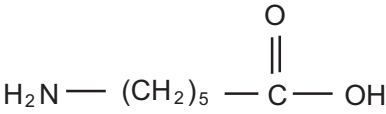
38 Which reaction(s) show(s) a large positive increase in entropy, ΔS ?

- 1 $[\text{Cr}(\text{H}_2\text{O})_6]^{3+} + 3 \text{ en} \rightarrow [\text{Cr}(\text{en})_3]^{3+} + 6 \text{ H}_2\text{O}$
- 2 $[\text{Cr}(\text{H}_2\text{O})_6]^{3+} + \text{EDTA}^{4-} \rightarrow [\text{Cr}(\text{EDTA})]^{-} + 6 \text{ H}_2\text{O}$
- 3 $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4 \text{ NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + 4 \text{ H}_2\text{O}$

A ☐ B ☐ C ☐ D ☐

[1]

39 Which organic compounds can be used as monomers to form polyamides?

1	
2	
3	

A ☐ B ☐ C ☐ D ☐

[1]

40 Which compound(s) react(s) with ethanoic anhydride, $(\text{CH}_3\text{CO})_2\text{O}$, to produce an ester?

- 1 CH_3CONH_2
- 2 $\text{C}_6\text{H}_5\text{OH}$
- 3 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

A ☐ B ☐ C ☐ D ☐

[1]

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